PRACTICE SHEET 2

Electric Potential • Electrostatic Potential Energy & Work • Electric Dipole • Basic Electrostatics / Mixed

Time Allowed: 60 Mins **Total Questions: 20**

Name: _____ Class/Sec: _____ Date: _____

Instructions: Attempt all questions. Each question carries one correct option unless marked *one or more than one option is correct*. Questions without options are numerical-value type: write the answer in the blank. The tag after each question shows its source examination. A self-assessment page follows; the answer key is provided as a separate sheet.

Q.1 The electrostatic potential on the surface of uniformly charged spherical shell of radius $R = 10$ cm is 120 V. The potential at the centre of shell, at a distance $r = 5$ cm from centre, and at a distance $r = 15$ cm from the centre of the shell respectively, are: *[JEE Main 2025, 3 Apr, Mor.]*

(A) 0 V, 120 V, 40 V (B) 120 V, 120 V, 80 V

(C) 40 V, 40 V, 80 V (D) 0 V, 0 V, 80 V

Q.2 An electric charge $10^{-6}\,\mu$C is placed at origin $(0, 0)$ m of X–Y co-ordinate system. Two points P and Q are situated at $(\sqrt{3}, \sqrt{3})$ m and $(\sqrt{6}, 0)$ m respectively. The potential difference between the points P and Q will be: *[JEE Main 2024, 27 Jan, Mor.]*

(A) $\sqrt{3}$ V (B) $\sqrt{6}$ V (C) 0 V (D) 3 V

Q.3 If the electric potential at any point (x, y, z) m in space is given by $V = 3x^2$ volt. The electric field at the point $(1, 0, 3)$ m will be: *[JEE Main 2022, 29 Jun, Eve.]*

(A) 3 Vm^{-1}, directed along positive x-axis.
(B) 3 Vm^{-1}, directed along negative x-axis.
(C) 6 Vm^{-1}, directed along positive x-axis.
(D) 6 Vm^{-1}, directed along negative x-axis.

Q.4 Concentric metallic hollow spheres of radii R and $4R$ hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference $V(R) - V(4R)$ is: *[JEE Main 2020, 3 Sep, Eve.]*

(A) $\dfrac{3Q_2}{4\pi\epsilon_0 R}$ (B) $\dfrac{3Q_1}{4\pi\epsilon_0 R}$

(C) $\dfrac{3Q_1}{16\pi\epsilon_0 R}$ (D) $\dfrac{Q_2}{4\pi\epsilon_0 R}$

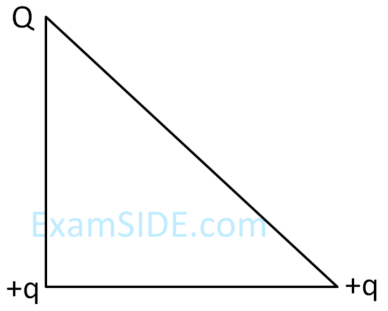
Q.5 Assume that an electric field $\vec{E} = 30x^2\hat{i}$ exists in space. Then the potential difference $V_A - V_O$, where V_O is the potential at the origin and V_A the potential at $x = 2$ m is: *[JEE Main 2014 (Offline)]*

(A) 120 J/C (B) −120 J/C (C) −80 J/C (D) 80 J/C

Q.6 Two point charges of 1 nC and 2 nC are placed at the two corners of equilateral triangle of side 3 cm. The work done in bringing a charge of 3 nC from infinity to the third corner of the triangle is _____ μJ. $\left(\dfrac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)$ *[JEE Main 2026, 28 Jan, Mor.]*

(A) 3.3 (B) 27 (C) 2.7 (D) 5.4

Q.7 Three charges Q, $+q$ and $+q$ are placed at the vertices of a right-angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is: *[JEE Main 2019, 11 Jan, Mor.]*



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(A) $\dfrac{-q}{1+\sqrt{2}}$ (B) $+q$

(C) $-2q$ (D) $\dfrac{-\sqrt{2}\,q}{\sqrt{2}+1}$

Q.8 Two points P and Q are maintained at the potentials of 10 V and -4 V, respectively. The work done in moving 100 electrons from P to Q is: *[AIEEE 2009]*

(A) 9.60×10^{-17} J (B) -2.24×10^{-16} J

(C) 2.24×10^{-16} J (D) -9.60×10^{-17} J

Q.9 Eight similar drops of mercury are maintained at 12 V each. All these spherical drops combine into a single big drop. The potential energy of bigger drop will be _____ E. Where E is the potential energy of a single smaller drop. *[JEE Main 2022, 30 Jun, Mor.]*

Q.10 The electrostatic potential due to an electric dipole at a distance r varies as: *[JEE Main 2024, 30 Jan, Mor.]*

(A) $\frac{1}{r^3}$ (B) $\frac{1}{r}$

(C) $\frac{1}{r^2}$ (D) r

Q.11 The electric field due to a short electric dipole at a large distance (r) from center of dipole on the equatorial plane varies with distance as: *[JEE Main 2023, 15 Apr, Mor.]*

(A) $\frac{1}{r^2}$ (B) $\frac{1}{r}$

(C) r (D) $\frac{1}{r^3}$

Q.12 Two charges each of magnitude 0.01 C and separated by a distance of 0.4 mm constitute an electric dipole. If the dipole is placed in an uniform electric field $\vec{E}$ of 10 dyne/C making 30° angle with $\vec{E}$, the magnitude of torque acting on dipole is: *[JEE Main 2023, 13 Apr, Mor.]*

(A) 4.0×10^{-10} Nm (B) 1.5×10^{-9} Nm

(C) 1.0×10^{-8} Nm (D) 2.0×10^{-10} Nm

Q.13 An electric dipole is formed by two equal and opposite charges q with separation d. The charges have same mass m. It is kept in a uniform electric field E. If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is: *[JEE Main 2019, 8 Apr, Eve.]*

(A) $\sqrt{\frac{qE}{md}}$ (B) $\sqrt{\frac{qE}{2md}}$

(C) $\sqrt{\frac{qE}{-2md}}$ (D) $\sqrt{\frac{2qE}{md}}$

Q.14 An electric field of 1000 V/m is applied to an electric dipole at angle of 45°. The value of electric dipole moment is 10^{-29} C·m. What is the potential energy of the electric dipole? *[JEE Main 2019, 11 Jan, Eve.]*

(A) -7×10^{-27} J (B) -9×10^{-20} J

(C) -10×10^{-29} J (D) -20×10^{-18} J

Q.15 An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience *[AIEEE 2006]*

(A) a translation force only in the direction of the field

(B) a translation force only in a direction normal to the direction of the field

(C) a torque as well as a translational force

(D) a torque only

Q.16 Three small identical bubbles of water having same charge on each coalesce to form a bigger bubble. Then the ratio of the potentials on one initial bubble and that on the resultant bigger bubble is: *[JEE Main 2026, 22 Jan, Eve.]*

(A) $1 : 3^{2/3}$ (B) $3^{2/3} : 1$

(C) $1 : 3^{1/3}$ (D) $1 : 2^{2/3}$

Q.17 Two identical positive charges Q each are fixed at a distance of $2a$ apart from each other. Another point charge q_0 with mass m is placed at midpoint between two fixed charges. For a small displacement along the line joining the fixed charges, the charge q_0 executes SHM. The time period of oscillation of charge q_0 will be: *[JEE Main 2022, 27 Jul, Mor.]*

(A) $\sqrt{\frac{4\pi^3\epsilon_0 ma^3}{q_0 Q}}$ (B) $\sqrt{\frac{q_0 Q}{4\pi^3\epsilon_0 ma^3}}$

(C) $\sqrt{\frac{2\pi^2\epsilon_0 ma^3}{q_0 Q}}$ (D) $\sqrt{\frac{8\pi^3\epsilon_0 ma^3}{q_0 Q}}$

Q.18 Two identical charged particles each having a mass 10 g and charge 2.0×10^{-7} C are placed on a horizontal table with a separation of L between them such that they stay in limiting equilibrium. If the coefficient of friction between each particle and the table is 0.25, find the value of L. [Use $g = 10$ ms^{-2}] *[JEE Main 2022, 24 Jun, Eve.]*

(A) 12 cm (B) 10 cm (C) 8 cm (D) 5 cm

Q.19 In free space, a particle A of charge $1\,\mu$C is held fixed at a point P. Another particle B of the same charge and mass $4\,\mu$g is kept at a distance of 1 mm from P. If B is released, then its velocity at a distance of 9 mm from P is: $\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}\right]$ *[JEE Main 2019, 10 Apr, Eve.]*

(A) 1.0 m/s (B) 6.32×10^4 m/s

(C) 2.0×10^3 m/s (D) 1.5×10^2 m/s

Q.20 A charged particle q is shot towards another charged particle Q which is fixed, with a speed v. It approaches Q upto a closest distance r and then returns. If q were given a speed of $2v$ the closest distance of approach would be *[AIEEE 2004]*

(A) $r/2$ (B) $2r$ (C) r (D) $r/4$

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Integrated Programme | Topic: Electrostatics – Potential, Potential Energy & Dipole

SELF-ASSESSMENT – PRACTICE SHEET 2

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 20	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____ % (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	**Q.20** ☐C ☐W ☐S

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake

☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

_____

_____

Topics / questions I need to revise:

_____

_____

My action plan before the next sheet:

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____